

MAE Purifier Defense vs Baseline ResNet-50 — Results

Numbers-and-results doc (not a paper draft). Defense = `MAE_defense_model`.

Setup

- **Defense (`MAE_defense_model`):** a **test-time input purifier** in front of an **unmodified, pretrained ResNet-50** — `prediction = resnet50( purify(x) )`. The purifier is a **frozen pretrained facebook/vit-mae-base** that masks a fraction of the 16×16 patches and reconstructs them, averaged over K random masks. **No training**; the ResNet-50 weights are never changed. ([models/mae_defense.py](#))
- `mask_ratio = 0.5`, `K = 8`, `MAE = facebook/vit-mae-base`.
- **Baseline:** the same pretrained torchvision ResNet-50, no purifier.
- **Adaptive evaluation (honest gradients).** The purifier is non-differentiable (runs under `no_grad`) and stochastic. To avoid a false "obfuscated gradients" result, logits uses a **BPDA straight-through estimator** (forward = purified image; backward = identity through the purifier). So **FGSM/PGD here are adaptive attacks against the defense**, not naive attacks on the bare classifier.
- **Protocol:** $\epsilon = 8/255$ L_∞ , seed = 42, 1000-image class-balanced ImageNet-val subset; PGD-20; Tesla T4. Same images/ ϵ /steps as the baselines.
- **■ Incomplete evaluation: AutoAttack and Square were skipped this session (compute).** They are the gradient-masking cross-checks and are pending a later Kaggle session. Read every robustness number below as **provisional** until they are run — see *Pending attacks*.

Headline

The purifier is **cheap** (clean accuracy 78.1% → **74.2%**, only −3.9 pp) but it **does not provide white-box adversarial robustness**. Under the **adaptive BPDA-PGD-20** attack it scores **1.3%**, essentially tied with the baseline's **0.4%** — both at floor. The earlier *non-adaptive* smoke test (attack the bare classifier, then purify) recovered ~31%; that signal **disappears once the attack is made adaptive** — a textbook obfuscated-gradients result.

Axis 1 — Gradient attacks (white-box, adaptive via BPDA)

Attack	Baseline R50	MAE defense	Δ
Clean	0.781	0.742	−0.039
FGSM ($\epsilon=8/255$)	0.408	0.414	+0.006
PGD-20 (BPDA-adaptive)	0.004	0.013	+0.009
AutoAttack	—	—	<i>pending</i>

Square	—	—	<i>pending</i>
--------	---	---	----------------

Read: FGSM accuracy (~41%) is **the same with or without the purifier** — that is ResNet-50's own single-step resistance, not a defense effect. Under the multi-step adaptive PGD both models collapse to the floor (1.3% = 13/1000 vs 0.4% = 4/1000); the +0.9 pp is 9 images at floor, **not** a robustness gain. **No white-box robustness.**

Axis 2 — Transfer attack (surrogate = bare ResNet-50, PGD-20)

Model	Clean	Transfer-robust	Drop	Fooling rate
Baseline R50 (white-box on self)	0.781	0.003	0.778	99.6%
MAE defense	0.742	0.053	0.689	92.9%

Read: the **one place the purifier measurably helps**. Adversarials crafted on the *bare* ResNet-50 partly fail to transfer through the purifier (5.3% survive vs 0.3%). Because the defense shares the ResNet-50 backbone with the surrogate, this is effectively a *same-backbone* transfer, so the +5 pp is a genuine — if small — disruption effect. Still mostly broken (92.9% fooling).

Axis 3 — Typographic attack (targeted, lowercase first-synonym)

Model	Clean	Robust	TASR	Fooling rate
Baseline R50	0.781	0.768	0.001	3.6%
MAE defense	0.742	0.724	0.000	5.8%

Read: no effect either way — ResNet-50 doesn't read printed words (TASR ≈ 0 for both). The defense is marginally **worse** (robust 72.4% vs 76.8%), tracking its lower clean accuracy. No win here (expected; typographic is a CLIP phenomenon).

Axis 4 — Common corruptions (severity 3, 15 types)

Model	Mean robust
Baseline R50	0.498
MAE defense	0.480

Mean is **−1.8 pp worse**, but the per-type breakdown shows a real, interpretable pattern — the purifier (a reconstruction model) **helps on noise, hurts elsewhere**:

C	B	M ^Δ _d	Corruption	Baseline	MAE def	Δ
---	---	-----------------------------	------------	----------	---------	---

gaussian_noise	0.721	0.686	-0.035
shot_noise	0.700	0.656	-0.044
glass_noise	0.447	0.413	-0.034
improved_elastic_transform	0.771	0.728	-0.043
fog	0.384	0.337	-0.047
frost	0.618	0.583	-0.035
jpeg_compression	0.660	0.649	-0.011
motion_blur	0.129	0.118	-0.011
pixelate	0.594	0.576	-0.018
snow	0.664	0.652	-0.012
zoom_blur	0.393	0.359	-0.034

Read: the purifier improves the three **noise** corruptions (gaussian/shot/glass) — consistent with it being a denoiser — but slightly degrades the blur/weather/clean types, so the mean nets out negative. A nice mechanism check, not a robustness win.

Where it wins / where it fails

Real (modest) effects: - **Transfer:** 5.3% vs 0.3% — partial disruption of transferred adversarials. - **Noise corruptions:** small gains on gaussian/shot/glass noise (it denoises).

Failures / costs: - **White-box PGD (adaptive/BPDA): no robustness** — 1.3%, floor, $\approx$ baseline. - **FGSM:** identical to baseline ($\sim 41\%$); purifier adds nothing. - **Typographic / corruptions mean:** slightly worse (clean-accuracy carryover). - **Clean accuracy:** -3.9 pp.

The obfuscated-gradients lesson (the actual finding)

This run is most useful as a **clean, adaptively-evaluated negative result:**

PGD-20 vs the MAE purifier	robust acc
Non-adaptive (attack bare R50, then purify) — mae_smoke	~ 0.31
Adaptive (BPDA straight-through) — this run	0.013

The 30-point gap between the two **is** the obfuscated-gradients pitfall, measured directly: a purifier that looks robust under a naive attack collapses to floor once the attacker differentiates through it. That contrast is an honest, teachable result on its own.

Pending attacks (and the expectation)

AutoAttack and Square were skipped (compute) and will run in a later session. Expectation, stated honestly up front: - **AutoAttack** (APGD $\geq$ PGD strength) attacks via the same BPDA path $\rightarrow$ it will **not rescue** the defense; expect $\approx$ floor, like PGD. - **Square** (gradient-free, queries the true stochastic pipeline) is the remaining check on whether the residual 1.3% is real robustness or just stochastic-masking noise. If Square also reads $\sim$ floor, the null-robustness verdict is confirmed.

The provisional verdict is therefore a **null adversarial-robustness result**; the pending attacks are expected to **confirm**, not overturn it.

Honest verdict

The MAE purifier is a **low-cost, model-agnostic front-end** that gives a small transfer-robustness and noise-corruption benefit, **but it does not beat the baseline on white-box adversarial robustness** — under an adaptive (BPDA) attack it is at floor, like the undefended model. As an adversarial *defense* it is, on the evidence so far, a **null result** — but a scientifically clean one, and the non-adaptive-vs-adaptive gap is a genuine finding worth reporting.

Reproducibility

- Seed 42; $\epsilon = 8/255 L_\infty$; PGD-20 (step 2/255); 1000-image class-balanced subset.
- Defense: facebook/vit-mae-base (frozen), mask_ratio=0.5, K=8, BPDA straight-through; ResNet-50 = torchvision DEFAULT weights (unmodified).
- Host: Tesla T4, torch 2.10.0+cu128, torchvision 0.25.0+cu128, transformers 5.0.0.
- Per-cell JSONs:
`results/{gradient,transfer,typographic,corruptions}/MAE_defense_model*.`
- **Not yet run:** AutoAttack, Square (gradient-masking cross-checks).